

Jeremy Devin

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Work Experience

Software Engineer, Web Platform & Agentic Tooling | Capital One

San Francisco, CA

ANGULAR, TYPESCRIPT, CLAUDE AGENT SDK, AWS BEDROCK

Aug 2021 - Present

- Leading development of an **agentic pipeline** (Claude Agent SDK, 5-agent architecture) to migrate 400+ legacy Discover web pages onto Capital One's platform, automating what was previously a fully manual, page-by-page process
- Refined the component generation agent routing LLM calls through AWS Bedrock, with **automated visual-regression scoring** (Playwright) validating migrated output against all 400+ source pages
- Optimized performance on site-wide navigation menu, reducing landing page load times by an average of 6%
- Piloted, documented, and rolled out an **AI-powered component development workflow**, reducing average dev time by 70%
- **Built and maintained Angular components** on capitalone.com, used by 80+ engineers and 200+ content managers
- Conducted **technical interviews** for internship and full-time engineering candidates

Software Engineer, Data Lineage | Capital One

San Francisco, CA

REACT, TYPESCRIPT, AWS

Aug 2020 - Aug 2021

- Designed and implemented UI for an internal **data lineage application**, enhancing data traceability within the company
- Modernized and refactored components to maximize cross-CMS compatibility and reduce developer friction
- Served as **Learning & Development lead** for the engineering rotational program council, organizing lunch-and-learns and building a shared library of onboarding resources

Projects

ARC-AGI Reasoning Agent

PYTHON, CASE-BASED REASONING, SEARCH

- Designed and implemented a hybrid agent combining **case-based reasoning** with generate-and-test/means-ends analysis to solve ARC-AGI abstract visual reasoning puzzles, composing across 50 grid-transformation operators
- Built an evaluation harness scoring agent generalization across held-out Training and Test sets, informing iterative refinement of the operator library and search strategy
- Authored an accompanying technical report formalizing the agent's architecture, reasoning strategy, and evaluation methods

Deep Learning for Lung Cancer Detection

PYTORCH, PYTHON, GOOGLE COLAB

- Developed custom CNNs for lung cancer detection, surpassing a published research benchmark with 95.19% accuracy.
- Engineered a minimal CNN, reducing model size to less than 1/30th of the original while still outperforming the benchmark.

Inclusion RVA

REACT, JAVASCRIPT, PYTHON

- Worked with Capital One Pro Bono and West Cary Group to build a full-stack web application, with an AWS Amplify-powered self-service auth portal, for search and discovery of minority-owned small businesses in the Richmond, Virginia area
- Won the Economic Impact Award, given to the Capital One Pro Bono team with the highest-valued economic contribution

Skills

Languages: TypeScript, Python, JavaScript, SQL

AI/Agentic: Claude Agent SDK, AWS Bedrock, multi-agent orchestration, evaluation design

Certifications: AWS Certified Solutions Architect – Associate (2019 and 2023)

Education

Georgia Institute of Technology

Atlanta, GA

M.S. IN COMPUTER SCIENCE (IN PROGRESS)

August 2024 - Present

- Specialization in Artificial Intelligence

Vanderbilt University

Nashville, TN

B.S. IN COMPUTER SCIENCE AND MATHEMATICS

Aug 2016 - May 2020

- Minors: Engineering Management, Scientific Computing